

Discussion 6: Symmetries and group theory – Solutions

Names:

Section:

In lecture, we defined the abstract properties of a group and showed that matrices M which preserved the lengths of vectors satisfied an additional special property, $M^T M = I$. In this discussion we will make these abstract considerations very concrete by looking at the specific examples of rotations in 2 and 3 dimensions.

Part 1: Rotations in 2 dimensions

Recall the general form of a rotation matrix from Discussion 3:

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \quad (1)$$

We will think of this as an active transformation which rotates vectors counterclockwise by the angle θ .

1.1. Verify that $R(\theta)^T = R(\theta)^{-1}$, in other words, that $R(\theta)$ is an orthogonal matrix.

$R(\theta)^T R(\theta) = \begin{pmatrix} \cos^2 \theta + \sin^2 \theta & -\sin \theta \cos \theta + \cos \theta \sin \theta \\ \cos \theta \sin \theta - \sin \theta \cos \theta & \cos^2 \theta + \sin^2 \theta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. Since $R(\theta)^T R(\theta) = I$, $R(\theta)^T$ is the inverse to $R(\theta)$. We could also have identified the inverse to $R(\theta)$ as $R(-\theta)$, which is rotation in the other direction.

1.2. Show by *explicit calculation* that the set of matrices $R(\theta)$ forms a group. To do this, you need to show:

- (a) Identity: there is some θ such that $R(\theta) = I$
- (b) Inverse: for any θ , there is a θ' such that $R(\theta') = R(\theta)^{-1}$
- (c) Closure: for any θ_1 and θ_2 , $R(\theta_1)R(\theta_2)$ corresponds to $R(\theta_3)$ for some θ_3 .

The identity element is $R(0) = I$. In 1.1 we identified the inverse element to $R(\theta)$ as $R(-\theta)$. To show closure, note that

$$\begin{aligned} R(\theta_1)R(\theta_2) &= \begin{pmatrix} \cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2 & -\cos \theta_1 \sin \theta_2 - \sin \theta_1 \cos \theta_2 \\ \sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2 & -\sin \theta_1 \sin \theta_2 + \cos \theta_1 \cos \theta_2 \end{pmatrix} \\ &= \begin{pmatrix} \cos(\theta_1 + \theta_2) & -\sin(\theta_1 + \theta_2) \\ \sin(\theta_1 + \theta_2) & \cos(\theta_1 + \theta_2) \end{pmatrix} \end{aligned}$$

So $\theta_3 = \theta_1 + \theta_2$.

- 1.3. The group of matrices $R(\theta)$ is given the name $\text{SO}(2)$, which means “special orthogonal group in 2 dimensions.” You already know what “orthogonal” means; “special” means that all the matrices have determinant equal to 1. Show by explicit calculation that $\det R(\theta) = 1$.

$$\det R(\theta) = \cos^2 \theta + \sin^2 \theta = 1.$$

- 1.4. Show by explicit calculation that $R(\theta_1)R(\theta_2) = R(\theta_2)R(\theta_1)$. The matrices in $\text{SO}(2)$ *commute*; groups that have this property are called *abelian*.

Multiplying the matrices in the reverse order gives the same matrix as we found in 1.2.

- 1.5. For a very small rotation angle ϵ , we can Taylor-expand the entries of $R(\epsilon)$ to first order in ϵ . Do the Taylor expansion of the sines and cosines to show that $R(\epsilon) = I + \epsilon X + \dots$, where

$$X = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \quad (2)$$

For $\theta = \epsilon$, we have $\cos \epsilon = 1 - \mathcal{O}(\epsilon^2)$ and $\sin \epsilon = \epsilon - \mathcal{O}(\epsilon^3)$. So to $\mathcal{O}(\epsilon)$, $R(\epsilon) = \begin{pmatrix} 1 & -\epsilon \\ \epsilon & 1 \end{pmatrix} = 1 + \epsilon X$.

- 1.6. Just as ordinary single-variable calculus teaches us that we can build up a finite integral from a bunch of infinitesimal rectangles, we can build up a finite rotation angle from a bunch of infinitesimal ones. This is why X is called the *generator* of $\text{SO}(2)$. Applying transformations one after the other corresponds to matrix multiplication: if we want an angle θ , we take $\epsilon = \theta/n$ and compute

$$M = \lim_{n \rightarrow \infty} [R(\epsilon)]^n = \lim_{n \rightarrow \infty} \left(I + \frac{\theta}{n} X \right)^n. \quad (3)$$

That looks a heck of a lot like one of the definitions of the exponential function! Pushing this analogy a little further, we can define the *matrix exponential* by its Taylor series expansion,

$$\exp(\theta X) \equiv \sum_{n=0}^{\infty} \frac{\theta^n}{n!} X^n. \quad (4)$$

Note that this is a *definition*, and does *not* correspond to simply taking the exponentials of each matrix component; the terms on the right-hand side are honest-to-goodness matrix multiplication. By definition, $X^0 = I$ for any matrix X . To finish the argument, show that

$$\exp(\theta X) = R(\theta), \quad (5)$$

in other words, we do indeed recover a finite transformation $R(\theta)$ by performing many infinitesimal ones. *Hint: the infinite power series is not as scary as it seems! Compute X^2 , X^3 , and X^4 , and look for a pattern. You should recognize the Taylor series for sine and cosine hiding inside the components of those matrices.*

Computing the first few powers, $X^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I$, and thus $X^3 = X(X^2) = X(-I) = -X$. Finally, $X^4 = X(X^3) = -X^2 = I$. The pattern is therefore that the even powers of X

are equal to $\pm I$, and the odd powers are equal to $\pm X$. We can write the exponential as

$$\begin{aligned}\exp(\theta X) &= I + \theta X - \frac{\theta^2}{2!}I - \frac{\theta^3}{3!}X + \frac{\theta^4}{4!}I + \dots \\ &= \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots\right)I + \left(\theta - \frac{\theta^3}{3!} + \dots\right)X \\ &= (\cos \theta)I + (\sin \theta)X \\ &= \begin{pmatrix} \cos \theta & 0 \\ 0 & \cos \theta \end{pmatrix} + \begin{pmatrix} 0 & -\sin \theta \\ \sin \theta & 0 \end{pmatrix} \\ &= \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}\end{aligned}$$

1.7. Finally, consider the matrix

$$F = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (6)$$

Show that $F^T F = I$, but that F does not correspond to any $R(\theta)$ we have studied so far. Draw a sketch to show how F transforms a vector (x, y) . It turns out that F is a member of a related group, $O(2)$, consisting of orthogonal matrices with determinant ± 1 . However, it can't be obtained by applying a series of infinitesimal transformations as in 1.6.

We can do this by ordinary matrix multiplication, but it's convenient to note a couple useful tricks: $F^T = F$, since the diagonal elements vanish, and for a diagonal matrix, matrix multiplication is equivalent to multiplying the elements on the diagonal. So $F^T F = \begin{pmatrix} (-1)^2 & 0 \\ 0 & (1)^2 \end{pmatrix} = I$. F does not correspond to $R(\theta)$ because it has determinant -1 while all of the $R(\theta)$ have determinant $+1$. An equivalent way of stating this is that $R(\theta)$ must have the two diagonal elements equal, but F does not. F takes (x, y) to $(-x, y)$, which is a reflection about the y -axis.

Part 2: Rotations in 3 dimensions

The study of rotations in 3 dimensions is very similar to the 2-dimensional case you studied in Part 1. There are two additional features, though: the rotation group $SO(3)$ is not abelian (you saw this already in Discussion 5 last week), and there are *three* generators, not just one. These facts are related and make the analysis a bit more complicated, but bear with it: studying rotations will help us better understand Lorentz transformations next week!

The group $SO(3)$ is the set of orthogonal 3×3 matrices with determinant 1. Three types of elements in $SO(3)$ are counterclockwise rotations about the x , y , and z axes, which are given by

$$R_x(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix}, R_y(\theta) = \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix}, R_z(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (7)$$

- 2.1. Verify that $R_x(\theta)^T R_x(\theta) = I$ (where now I is the 3×3 identity matrix), and likewise for $R_y(\theta)$ and $R_z(\theta)$.¹

At this point you should be fairly comfortable with the mechanics of matrix multiplication – the only difference compared to Part 1, problem 1.1 is the presence of the 1 on the diagonal, which ensures that the matrix multiplication gives $\cos^2 \theta + \sin^2 \theta = 1$ for the other diagonal entries and $\cos \theta \sin \theta - \sin \theta \cos \theta = 0$ for the off-diagonal entries.

- 2.2. The minus sign is in a different place for R_y because we are working in a right-handed coordinate system, with counterclockwise rotations defined by the right-hand rule: if your right thumb points along the axis of rotation in the positive direction, your right hand curls in the direction of positive θ . Draw a sketch to convince yourself the sign is in the right place for R_y .

A right-handed rotation should take a vector along the z -axis towards the positive x -axis because $\hat{z} \times \hat{x} = \hat{y}$. Acting on $(0, 0, 1)$ with $R_y(\theta)$ gives $(\sin \theta, 0, \cos \theta)$, which has a positive x -component for $0 < \theta < \pi$.

- 2.3. Check that the matrices $R_x(\theta)$, $R_y(\theta)$, and $R_z(\theta)$ each form a group by themselves, using the same method as in question 1.2 of part 1 above.

The identity elements are $R_x(0)$, $R_y(0)$, and $R_z(0)$ (all of which are equal to the 3×3 identity matrix I). The inverses are $R_x(-\theta)$, $R_y(-\theta)$, and $R_z(-\theta)$. Multiplication gives $R_x(\theta_1)R_x(\theta_2) = R_x(\theta_1 + \theta_2)$ by an identical calculation to question 1.2, and likewise for R_y and R_z .

- 2.4. Taylor-expand $R_x(\epsilon)$, $R_y(\epsilon)$ and $R_z(\epsilon)$ as in question 1.5 to find $R_x(\epsilon) \approx I + \epsilon J_x$, $R_y(\epsilon) \approx I + \epsilon J_y$, and $R_z(\epsilon) \approx I + \epsilon J_z$, where

$$J_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad J_y = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad J_z = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (8)$$

The solution is identical to question 1.5 of part 1, with the sines and cosines placed in their appropriate positions in the 3×3 matrix. In particular, we recognize X as the lower-right 2×2 block of J_x and the upper-left 2×2 block of J_z , with the different placement and sign of J_y arising from the sign difference in R_y from 2.2.

- 2.5. Compute $(J_x)^T J_x$. Note that it's not equal to I ! The generators are not actually elements of the group they came from: they belong to a different mathematical structure called a *Lie* (pronounced “Lee”) *algebra*.

$(J_x)^T J_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. The lower 2×2 block is the 2×2 identity matrix, but the 0 in the first diagonal entry means that this matrix is not the 3×3 identity matrix.

¹The determinant of each of these matrices is indeed equal to 1, though checking this is a little more work because the determinant formula for matrices larger than 2×2 is a bit more complicated.

- 2.6. The fact that rotations don't commute (as you found in last week's discussion) is reflected in the *commutator* of the corresponding generators. The commutator is defined as $[X, Y] \equiv XY - YX$; in other words, multiply the matrices, multiply them in the reverse order, and subtract the result. Compute $[J_x, J_y]$ and show that it is equal to J_z .

$$\begin{aligned}
 [J_x, J_y] &= J_x J_y - J_y J_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \\
 &= \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\
 &= \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\
 &= J_z
 \end{aligned}$$

Computing the rest of the commutators (an exercise you are welcome to try if you have extra time) yields the *commutation relations*

$$[J_x, J_y] = J_z, \quad [J_y, J_z] = J_x, \quad [J_z, J_x] = J_y. \quad (9)$$

These are the same relations that define the cross-product of unit vectors for a right-handed coordinate system! ($\hat{x} \times \hat{y} = \hat{z}$, etc.) This is yet another manifestation of the relation between the group $\text{SO}(3)$ and rotations.² But there's more: the mathematics of group theory actually tells us that since the commutation relations are *closed* (meaning the commutator of any two generators yields another generator), *any* finite rotation about *any* axis can be built up through a combination of infinitesimal rotations about the three coordinate axes. In other words, we have shown in another way that $\text{SO}(3)$ is a group!

The reason this is so powerful is that it's not immediately obvious that, for example, $R_x(\alpha)R_y(\beta)$ is still a rotation. But by Taylor expanding, its generator can be seen to be a combination of J_x and J_y , so it must be an element of $\text{SO}(3)$. We will see next week how this helps us understand the effect of Lorentz boosts performed along different axes.

²You will see these commutation relations again in quantum mechanics: when you do, remember that they have their origin in infinitesimal rotations!